Data retention and deletion

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

One complete retention-policy and copy-verification record.

Keep policy version, age snapshot, listed stores, due IDs, residuals, allowed audit fields and verified scope together.

Keep one habit: Recall → W1–W3/W5 → independent 4-day case → project copy inventory → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Sandboxing and resource limits. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from External side effects and recovery. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Planning and action selection. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- Retention and deletion cover all declared stores, including caches and logs. Removing a visible row may leave copies elsewhere.

- Toy policy retains session content for 7 days and deletes it when age is at least 7. Audit metadata may remain for 30 days without content. Track primary storage, cache and backup policy separately.

Sandboxing and resource limits

> **My remembered idea, example and correction**
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External side effects and recovery

> **My remembered idea, example and correction**
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Planning and action selection

> **My remembered idea, example and correction**
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Lesson reference for workbook practice

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Main fictional policy removes session content at age at least 7 days and permits content-free audit metadata under a separate 30-day lifetime. At the supplied snapshot S6 is age 6, S7 age 7 and S9 age 9; primary/cache initially contain all three. Keep S6 and remove S7/S9 from both stores: four modeled copy removals. Main backup expiry is unverified, so do not claim global deletion.

  New audit events contain only ID/action/location/event age 0; audit age 29 is retained and age 30 is due under its own clock. Separate picture covers primary and the T2 backup: retain T1 at age 6, remove both T2 copies at age 7. A failed backup removal leaves scoped deletion incomplete. A separate retry can clear the remaining copy without overwriting its failed report.

  Empty inventories or listed copies with missing ages do not justify success. The code transforms copied fictional dictionaries; it deletes no real records and provides no secure-erasure guarantee. Preserve only policy-permitted metadata without expired content.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main fictional policy removes session content at age at least 7 days and permits content-free audit metadata under a separate 30-day lifetime. At the supplied snapshot S6 is age 6, S7 age 7 and S9 age 9; primary/cache initially contain all three. Keep S6 and remove S7/S9 from both stores: four modeled copy removals. Main backup expiry is unverified, so do not claim global deletion.

  New audit events contain only ID/action/location/event age 0; audit age 29 is retained and age 30 is due under its own clock. Separate picture covers primary and the T2 backup: retain T1 at age 6, remove both T2 copies at age 7. A failed backup removal leaves scoped deletion incomplete. A separate retry can clear the remaining copy without overwriting its failed report.

  Empty inventories or listed copies with missing ages do not justify success. The code transforms copied fictional dictionaries; it deletes no real records and provides no secure-erasure guarantee. Preserve only policy-permitted metadata without expired content.

- Record ages are 6, 7, 9 days. Cache contains copies of all three. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main fictional policy removes session content at age at least 7 days and permits content-free audit metadata under a separate 30-day lifetime. At the supplied snapshot S6 is age 6, S7 age 7 and S9 age 9; primary/cache initially contain all three. Keep S6 and remove S7/S9 from both stores: four modeled copy removals. Main backup expiry is unverified, so do not claim global deletion.

  New audit events contain only ID/action/location/event age 0; audit age 29 is retained and age 30 is due under its own clock. Separate picture covers primary and the T2 backup: retain T1 at age 6, remove both T2 copies at age 7. A failed backup removal leaves scoped deletion incomplete. A separate retry can clear the remaining copy without overwriting its failed report.

  Empty inventories or listed copies with missing ages do not justify success. The code transforms copied fictional dictionaries; it deletes no real records and provides no secure-erasure guarantee. Preserve only policy-permitted metadata without expired content.

- A report omits this qualification: This fictional retention exercise is not a statement of legal retention requirements. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main fictional policy removes session content at age at least 7 days and permits content-free audit metadata under a separate 30-day lifetime. At the supplied snapshot S6 is age 6, S7 age 7 and S9 age 9; primary/cache initially contain all three. Keep S6 and remove S7/S9 from both stores: four modeled copy removals. Main backup expiry is unverified, so do not claim global deletion.

  New audit events contain only ID/action/location/event age 0; audit age 29 is retained and age 30 is due under its own clock. Separate picture covers primary and the T2 backup: retain T1 at age 6, remove both T2 copies at age 7. A failed backup removal leaves scoped deletion incomplete. A separate retry can clear the remaining copy without overwriting its failed report.

  Empty inventories or listed copies with missing ages do not justify success. The code transforms copied fictional dictionaries; it deletes no real records and provides no secure-erasure guarantee. Preserve only policy-permitted metadata without expired content.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Fictional retention policy deletes records at age >=4 days. X is3 days, Y4, Z5. A copy of Y remains in a temporary log after its main record is deleted. Evaluate compliance.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Changed independent case

My independent explanation, trace and limits

MAIN LESSON · SECTION 10

Project · Keep the service evidence

> Preserve the contract, inputs, observed outcomes and limits.

Complete task and evidence

- Add a labelled data retention and deletion **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of data retention and deletion.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for Data retention and deletion. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for Data retention and deletion. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Retention is5 days, inclusive **deletion boundary**. Ages 4, 5, 6; a log stores complete content. Trace required action.

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task’s inputs,rule and response together.

Complete task and evidence

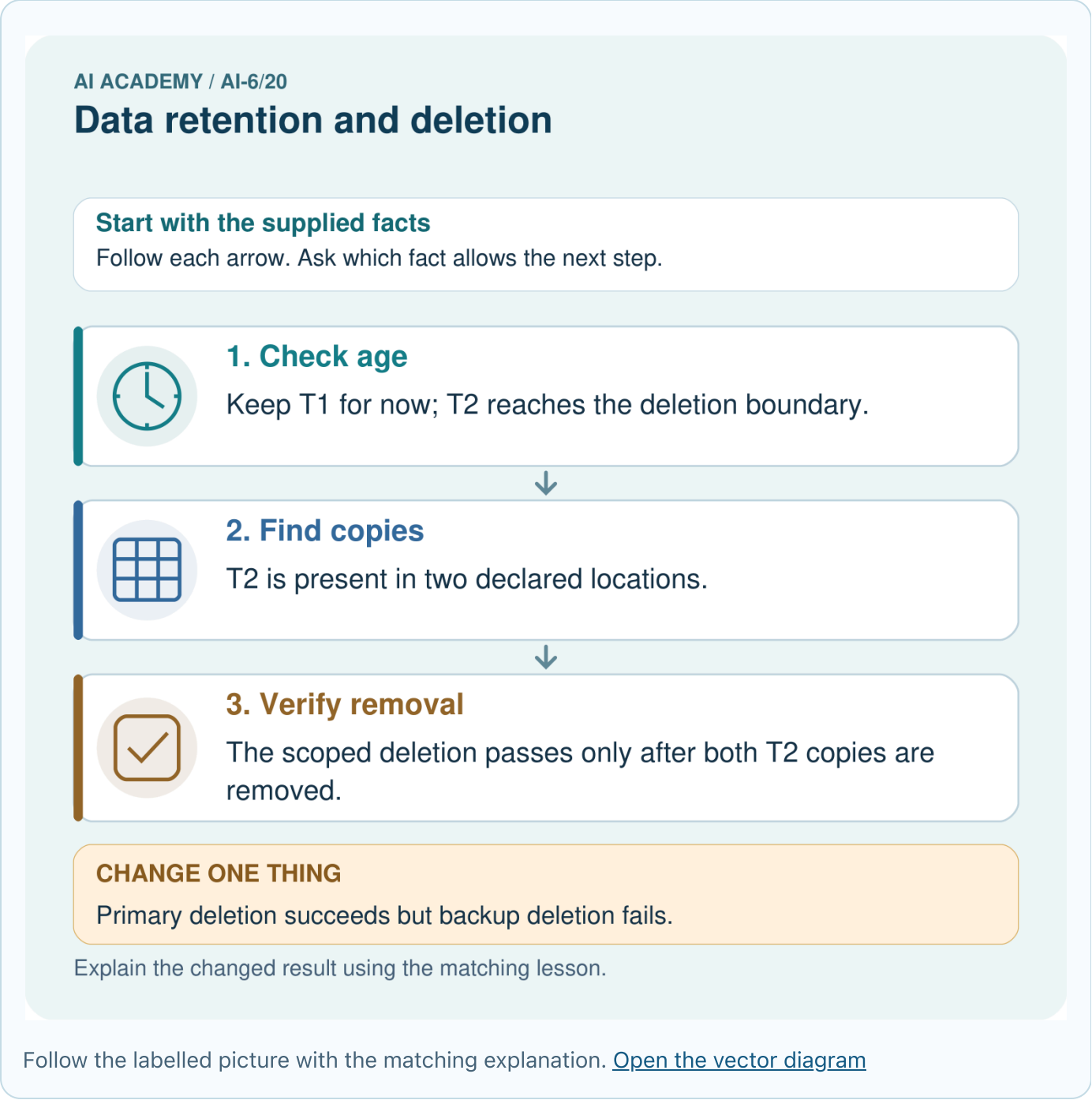
- Delete every listed copy. A fictional policy deletes transcripts at age seven days or older, including listed backups.

- T1 is six days old. T2 is seven days old with one backup. Scope lists primary storage and that backup only.

- Apply the seven-day boundary to T1 and T2. How many T2 locations need verification, and may deletion be called complete if its backup remains?

- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.



Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ **Reference · Project milestone**

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Grounded Learning Service
- Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.
- For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Delete every listed copy. A fictional policy deletes transcripts at age seven days or older, including listed backups.
- Which labelled step matches this detail? The scoped deletion passes only after both T2 copies are removed.
- 1. Check age
- 2. Find copies

- 3. Verify removal

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Verify removal The scoped deletion passes only after both T2 copies are removed. Retention requires an age rule and an inventory.

Deleting the visible copy alone does not demonstrate removal of listed backups.

Changed-case question and optional hint

- Changed-case practice: Primary deletion succeeds but backup deletion fails. Which conclusion is supported by the supplied facts?

- 1. Record incomplete deletion and follow the retry or escalation procedure.

- 2. We can decide the result without examining the changed **input** or the procedure.

- 3. This one example guarantees correct results for every future **input**.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Record incomplete deletion and follow the retry or escalation procedure. One required copy remains in the declared scope.

Repair and retain the record

Review Sandboxing and resource limits (ai-6/19). Goal: Explain the **mechanism** and terminology of sandboxing and resource limits. Short practice: Check A against both resource limits and B against network isolation.

Why must A stop after three seconds even if its **output** remains only 2 MB?. Return to this week and explain what you changed.

Review External side effects and recovery (ai-6/17). Goal: Explain the **mechanism** and terminology of external side effects and recovery. Short practice: Distinguish a missing printer acknowledgement from confirmed failure.

What should happen if J6 is complete, and what **evidence** changes when J6 is reliably reported absent?. Return to this week and explain what you changed.

Review Planning and action selection (ai-6/14). Goal: Explain the **mechanism** and terminology of planning and action selection. Short practice: Choose a permitted plan when no approved tower-height card exists.

What new answer becomes supported after a card records 24 metres, and why could the bot not invent it earlier?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

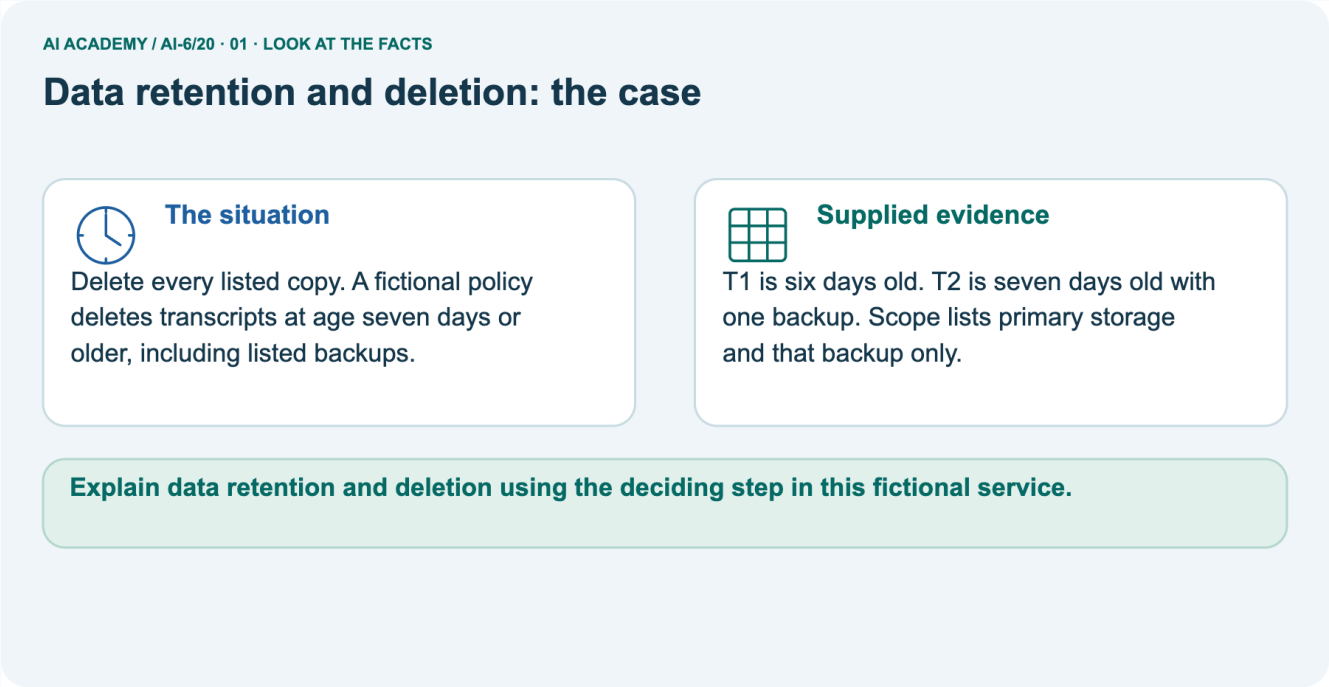
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Data retention and deletion: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

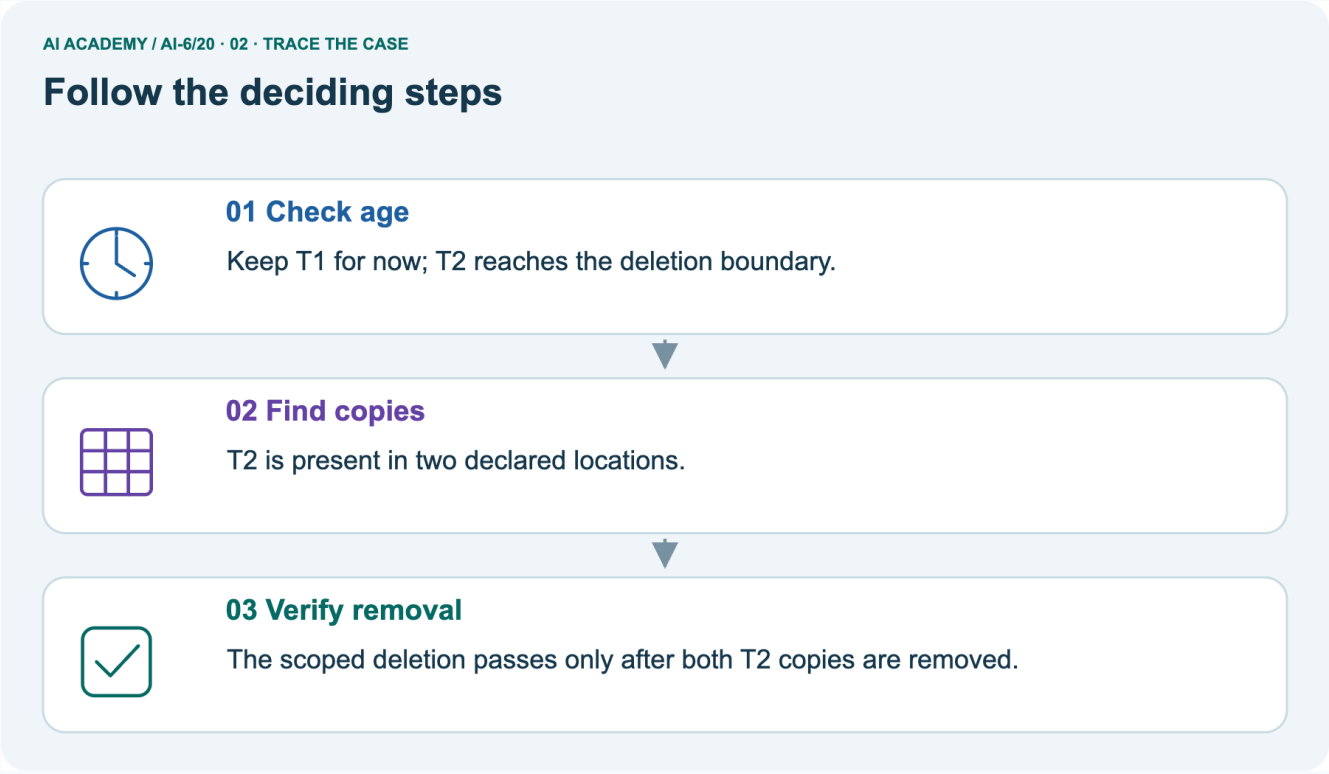
Delete every listed copy. A fictional policy deletes transcripts at age seven days or older, including listed backups. Supplied facts: T1 is six days old. T2 is seven days old with one backup.

Scope lists primary storage and that backup only.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



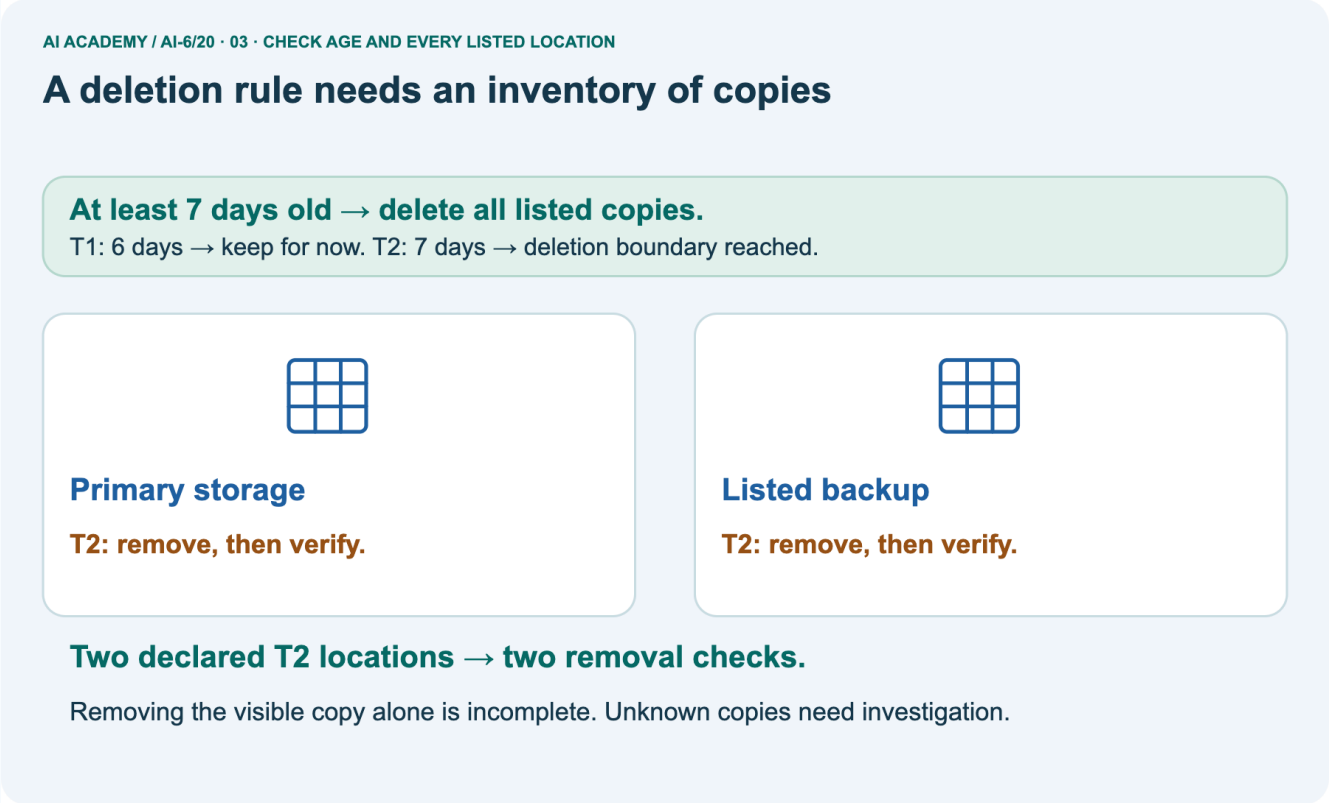
Follow the labelled picture with the matching explanation. Open the vector diagram

Check age: Keep T1 for now; T2 reaches the **deletion boundary**. → Find copies: T2 is present in two declared locations. → Verify removal: The scoped deletion passes only after both T2 copies are removed.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. A deletion rule needs an inventory of copies



Follow the labelled picture with the matching explanation. Open the vector diagram

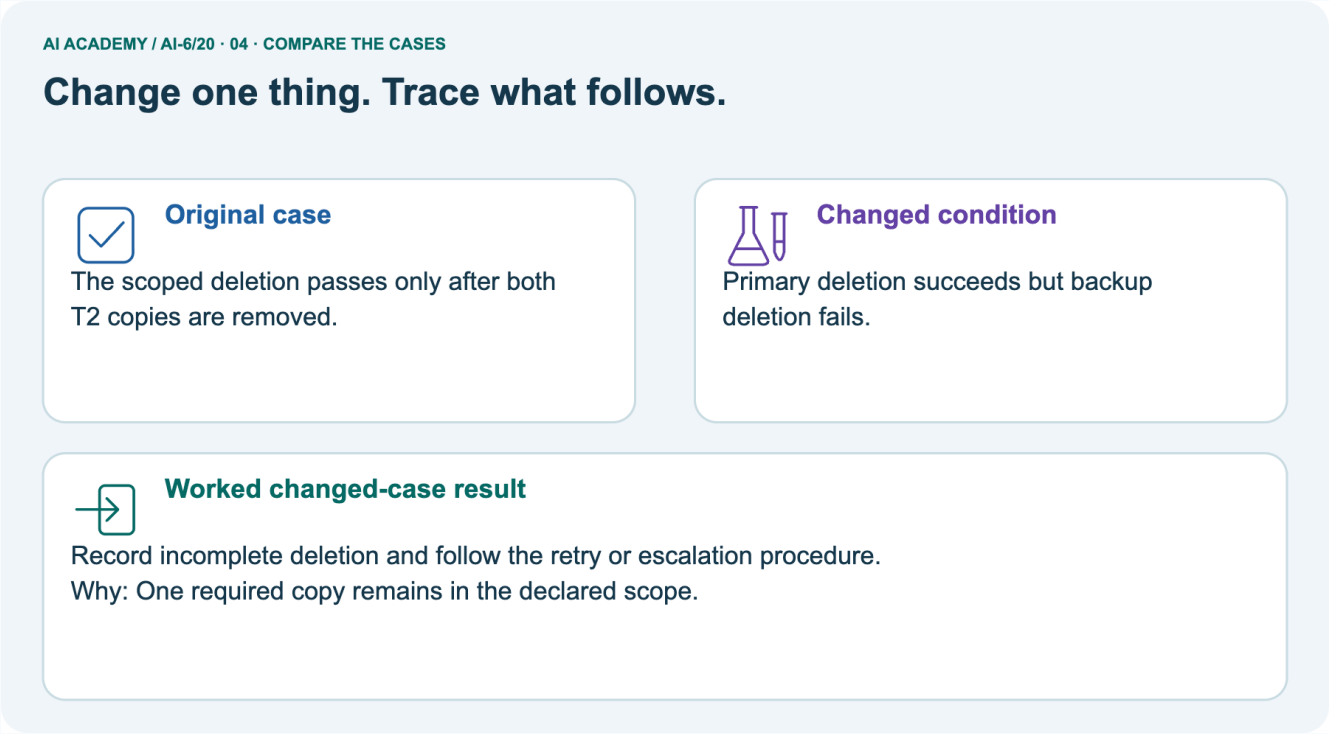
The toy retention boundary is seven days or older. Keep six-day T1 for now. Seven-day T2 appears in primary storage and one listed backup; both locations need removal verification.

This diagram specifies required checks, not an actual deletion run.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

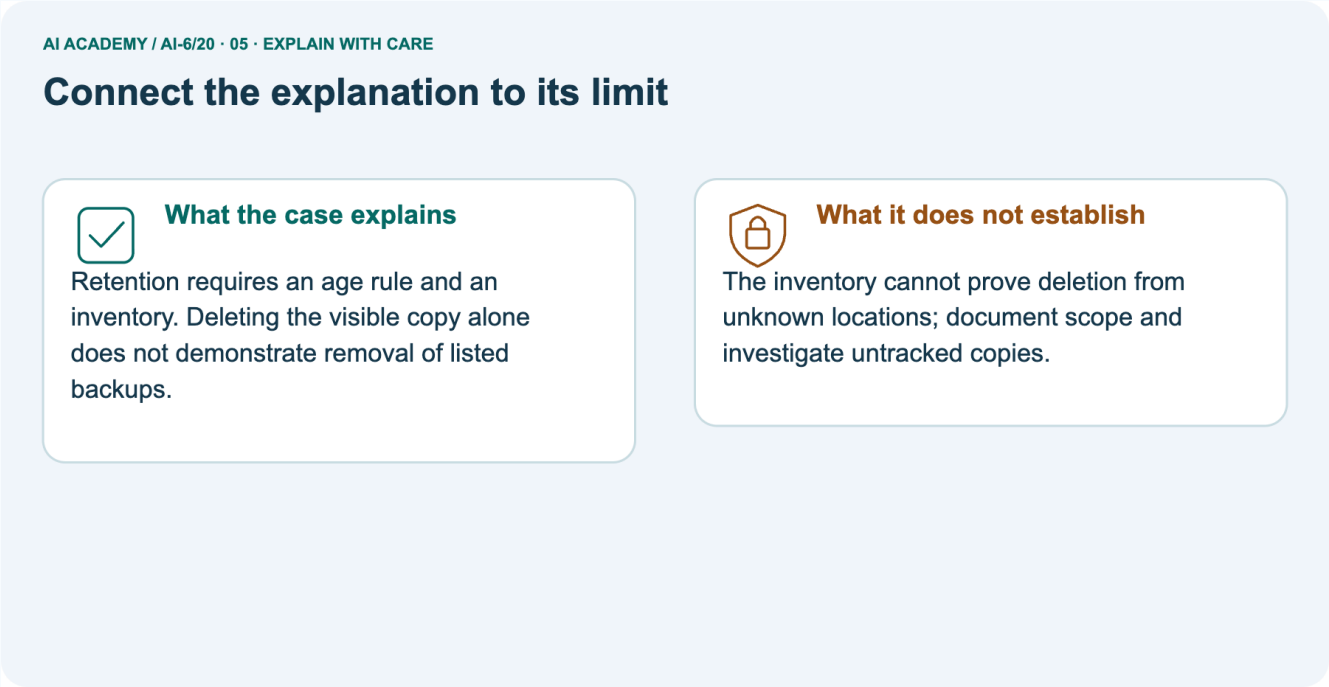
Original case: The scoped deletion passes only after both T2 copies are removed. Changed condition: Primary deletion succeeds but backup deletion fails.

Expected result: Record incomplete deletion and follow the retry or escalation procedure. Reason: One required copy remains in the declared scope.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

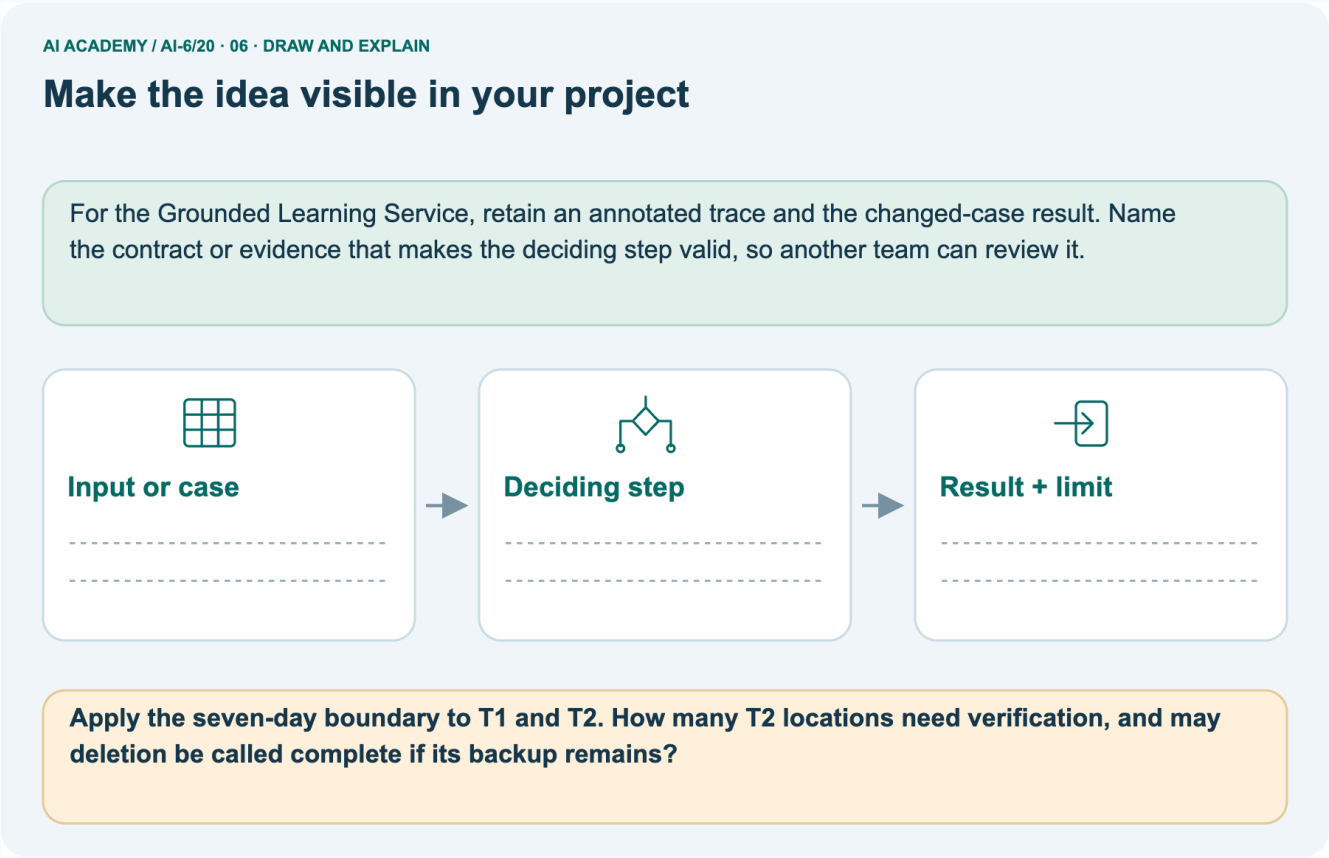
Retention requires an age rule and an inventory. Deleting the visible copy alone does not demonstrate removal of listed backups.

Limit: The inventory cannot prove deletion from unknown locations; document scope and investigate untracked copies.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Apply the seven-day boundary to T1 and T2. How many T2 locations need verification, and may deletion be called complete if its backup remains?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Data retention and deletion: the case

02. Follow the deciding steps

03. A deletion rule needs an inventory of copies

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.